Conventional MoE

Learn independent FFN weights.

Inputs

Positional encoding

Decoder block

Layer norm

Masked self-attention

+

Layer norm

Router top-2

Expert 1 W1

Expert 2 W2

Expert 3 W3

Expert 4 W4

Trainable FFN

Trainable FFN

Trainable FFN

Trainable FFN

+

Weighted sum

+

Remaining decoder blocks

Outputs

MASKerade

Learn masks. Freeze FFN weights.

Inputs

Positional encoding

Decoder block

Layer norm

Masked self-attention

+

Layer norm

Router top-2

Frozen W

Mask 1

Mask 2

Mask 3

Mask 4

Masked FFN

Masked FFN

Masked FFN

Masked FFN

+

Weighted sum

+

Remaining decoder blocks

Outputs